Development of Sustainable Education Indicators for Measuring a Green Campus

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ABSTRACT
Higher Education Institutions (HEIs) play a crucial role in a nation's development, making it essential for their growth to align with sustainable and environmentally friendly practices. Considering their fundamental role in teaching students at several levels, HEIs should exemplify sustainability. This study seeks to integrate educational indicators into a sustainable assessment tool (SAT) to evaluate green HEIs in Malaysia, addressing the urgent demand for sustainable development and the existing deficiency of green HEIs in the country. Although Malaysia has developed various SATs, none are suitable for evaluating green HEIs due to the lack of education-related indicators. To accomplish the research aim, this study has two objectives: to analyze potential education indicators and to propose suitable education indicators. As far as the authors are aware, the Ministry of Higher Education (MOHE) in Malaysia has not introduced any specific guidelines for incorporating sustainable education criteria into HEIs. The study employs a qualitative and quantitative data collection method via a questionnaire survey in 9 selected public HEIs in Malaysia, targeting academicians from specific faculties to validate these education indicators. The questionnaire survey results were analyzed using descriptive analysis with the aid of SPSS software, and the results have identified 30 important education indicators that should be considered in measuring sustainable HEIs. The top five indicators were identified as: (1) funding for program development, (2) funding for research and innovation, (3) funding for training, (4) sustainable courses, and (5) sustainable research. The findings will initially support the MOHE, policymakers, and green associations in creating a dedicated evaluation tool for assessing the green campus and guide other Malaysian HEIs in their journey toward becoming green campuses.

KEYWORDS: Green campus, Green higher education institutions, Education indicators, Sustainable development education, Developing countries

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1. INTRODUCTION

Higher education institutions (HEIs) play a crucial role in national development and should lead the way in expanding and implementing sustainable practices, given their substantial environmental impact (Bautista-Puig & Sanz-Casado, 2021; Tshivhase & Bisschoff, 2023). HEIs are increasingly crucial in advancing sustainability (Howarth et al., 2023) since they are considered 'small cities' (Anthony Jnr, 2021) due to the large community and campuses (Santa et al., 2019), a strong sense of social responsibility, and the vital role in the development of social behaviors (Heravi et al., 2021; Sepasi et al., 2019). Adopting sustainable practices in HEIs has been a longstanding global initiative since the 1990s (Rebich-Hespanha & Bales, 2023). The HEIs worldwide are taking on a significant role, not only in teaching and research, but also in advocating for sustainability beyond their institutional confines. Every HEI must prioritize sustainability, as this is no longer optional. Since HEIs can effectively spread impactful ideas and concepts to society, they play a key role in promoting sustainable thinking and increasing awareness of the importance. This is achieved not only through curricula and academic research but also through implementing positive environmental practices in educational buildings, aimed at reducing adverse impacts on the environment (Abdalla et al., 2020).

Measuring each achievement or goal against a standard is a common practice to ensure its quality. Assessment is one of the methods used to measure something's or someone's abilities towards a certain part, while assessment tools refer to the techniques used to do the measurement. Sustainability assessment tools (SATs) have been applied across various fields and industries to enhance decision-making at the policy, program, and project levels (Coteur et al., 2020). SATs are considered one of the initiatives towards the path of sustainability, and SATs have become the main elements in measuring sustainability (Du et al., 2020; Freidenfelds et al., 2018; Leal Filho et al., 2018).

In Malaysia, the Minister of Higher Education (MOHE) is responsible for the development of a better HEIs ecosystem. HEIs refer to a university, a university college, a university branch, a college or polytechnic, and community colleges and include both public and private institutions. The government funds public universities, while private universities are universities that are established by financially sound corporations. On the other hand, polytechnics and community colleges are established to train technical assistants and technicians in various engineering fields. The number of HEIs in Malaysia has also witnessed tremendous growth in the last two decades. There are more than 595 universities with 3.18 million students by the end of 2021. According to Fuchs et al. (2020), HEIs must be transformed into 'green campuses' to foster a community that is concerned with environmental issues. Green or sustainable HEIs are no longer an option. However, only 4.7% of HEIs in Malaysia were rated as sustainable based on the Global ranking UI Greenmetric. The number is extremely low and should not have happened because the size of HEI is similar to a 'small city' with a huge population. One of the reasons is due to the absence of sustainable criteria pertaining to the HEIs and the lack of indicators and initiatives to measure green HEIs (Adenle et al., 2020; Isa et al., 2021). Even though Malaysia has developed many SATs to evaluate sustainability, none of the tools were designed specifically to measure green HEIs. This is happening because none of the tools incorporates education criteria in evaluating green HEIs. This statement is supported by Rosa et al. (2024), whereby most sustainable tools do not mainly cover academic indicators in measuring sustainable HEIs. It should not have happened since the main role of HEIs is to educate youth and prepare future generations by providing adequate opportunities to learn about recent developments and future needs for sustainable societies (Habib et al., 2021). Due to this situation, this research aims to explore the education indicators that shall be included in sustainable assessment tools in evaluating sustainable HEIs. The development of these indicators addresses a key research gap and contributes to the novelty of this study. Education is seen as a key driver in shaping thoughts and actions, playing a crucial role in increasing public awareness of the consequences and impacts of unsustainable practices on society (Menon & Suresh, 2022; Thondhlana & Nkosi, 2024). Specifically, this study seeks to address two objectives:

I. Identifying important education indicators in developing sustainable HEIs.
II. Developing the education indicators for measuring sustainable HEIs in Malaysia.

2. LITERATURE REVIEW

2.1 Green Campuses, Criteria and Indicators

Green HEIs and Sustainable HEIs are regarded as similar based on the study by Anthony Jnr (2021). However, many experts argue that a Green Campus is a component of a Sustainable Campus, with the latter encompassing a broader scope. Sustainable HEIs are defined by Aleixo et al. (2018) and Dawodu et al. (2022) as an educational institution that grants academic degrees and focuses on minimizing the negative environmental, economic, societal, and health impacts generated by its resource use. This is achieved through its functions of teaching, outreach, partnerships, and stewardship, with the goal of helping society transition to a sustainable lifestyle, either on a regional or university level.

Meanwhile, Atici et al. (2021) described sustainable HEIs as institutions concerned with the environment and research, enhancing their campus setting and infrastructure to be more sustainable, and ensure their curricula covering environmental and sustainability courses. Findler et al. (2019) and Biancardi et al. (2023) claimed sustainable HEIs can be reached through education and research, fostering positive behaviors, and promoting community engagement, while Fissi et al. (2021) highlighted that the elements of sustainable HEIs shall consist of (1) curriculum, (2) research, (3) campus operation, (4) community engagement, and (5) teaching. Sustainable HEI is also known as institutions that can minimize negative impacts on the environment, economy, society, and well-being (Khovrak, 2020; Rinawiyanti et al., 2023). It is evident that most experts consider education, including curriculum and research, as a key criterion in developing a Green Campus. Solely focusing on environmental protection is not sufficient to make HEIs truly sustainable. **Table 1** presents other interpretations of green HEIs.

2.2 Tools for Measuring Green HEIs

Achieving a goal is challenging to measure without a proper assessment, as it provides the necessary framework to evaluate progress and success. Assessment involves evaluating someone's or something's performance based on specific criteria or objectives. It can be in the form of checklists, rubrics, multiple-choice questions, reports, etc. The Sustainable Assessment Tool (SAT) is one of the initiatives towards the path of sustainability and has become a crucial element in sustainable

Table 1: *Other definitions of green HEIs*

Green HEIs	Author/ year
Improving campus setting and infrastructure, and sustainable curriculum	(Atici et al., 2021)
Green HEIs can be achieved through building behaviors and community sharing	(Biancardi et al., 2023; Findler et al., 2019)
Green HEIs shall consist of elements which are campus operations, teaching, research, and community engagement	(Fissi et al., 2021)
Green HEIs minimize negative impacts on the environment, economy, society, and well-being	(Krskova et al., 2021; Rinawiyanti et al., 2023)
HEIs are created to promote the minimization of adverse environmental, economic, social, and health effects and fulfil their teaching, research, outreach, partnership, and stewardship functions.	(Aleixo et al., 2018; Dawodu et al., 2022)
Green HEIs should have sustainable campus operations, sustainable research, public outreach, cooperation between universities, sustainable curricula, and green reporting	(Freidenfelds et al., 2018)
Green campus aspects are governance, education, research, and engagement.	(Du et al., 2020)
Green campus resources and curricula, sustainable educational experience, sustainable administration and management of material resources, and sustainable governance.	(Gutiérrez-Mijares et al., 2023)
Green HEIs incorporates curriculum, operations, community outreach programs, and research.	(Menon & Suresh, 2022)

development (Du et al., 2023; Findler et al., 2019; Husaini et al., 2018; Parvez & Agrawal, 2019). SAT is a common practice not only in measuring sustainability but also used for products, policy and institutional appraisal. It can also be applied to countries, regions, states, cities, universities, or the planning and development of documents (Gomez & Yin, 2019). Using SATs for HEIs is not recent, as their implementation and adoption in various countries dates back to the 1990s, and the trend is rising (Husaini et al., 2018). This trend is evidenced by the proliferation of SATs developed over the years. Research conducted by Du et al. (2020) identified 40 SATs globally, indicating a growing interest in sustainable HEIs since their inception. Additionally, various studies have examined varying numbers of SATs for HEIs, further underscoring the increasing attention and research dedicated to this area (Dawodu et al., 2022; Parvez & Agrawal, 2019; Ribeiro et al., 2021; Zainordin & Ismail, 2018). Recently, the UI Greenmetric, a popular tool, has significantly increased the number of sustainable HEIs worldwide. This is evident from the rising number of sustainable HEIs since their establishment (**Figure 1**). However, measuring green HEIs remains complex and challenging due to the diverse practices and cultures in different regions.

2.3 Conceptual Framework

A framework for this research was formulated as per *Figure* 2. A conceptual framework is where the gaps of study were identified and where the study methodology was outlined (Varpio et al., 2020). The conceptual framework also helps the researcher to explain the contribution and novelty of the study. The framework encompasses two principal elements: the independent and dependent variables. In this study, the education criteria are conceptualized as

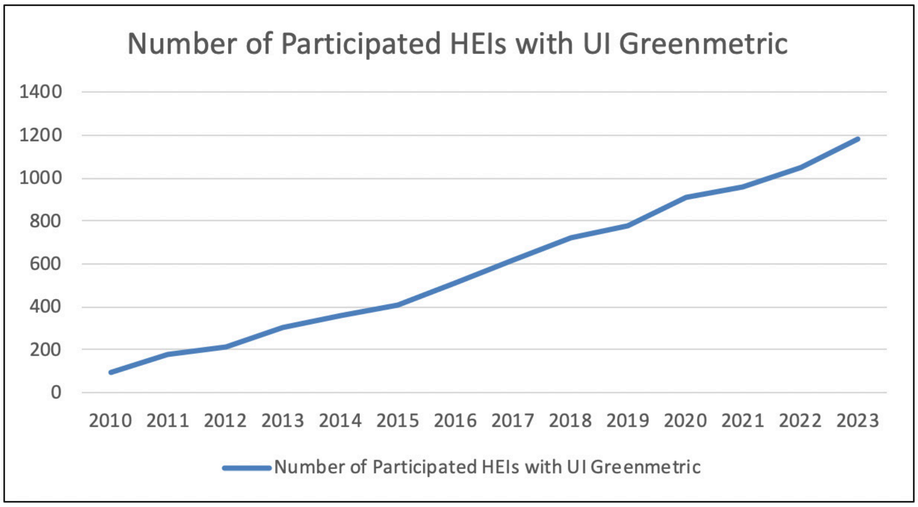

Figure 1: *Number of participating HEIs with UI Greenmetric since its establishment. Source: (Greenmetric, 2023)*

the independent variable, which the author seeks to develop and integrate into the Sustainability SATs for evaluating sustainable HEIs in Malaysia. The dependent variable is delineated through the corresponding main and sub-indicators derived from the education criteria.

3. MATERIALS AND METHODOLOGY

A mixed-method approach was applied in this study, utilizing both qualitative and quantitative data through secondary data collection and a questionnaire survey. There were two phases of the data collection, namely Phase 1: Identification of education indicators through a secondary data collection method, and Phase 2: Validation of Education Indicators through a Questionnaire survey.

3.1 Stage 1 Data Collection for Identification of Sustainable Education Indicators for Measuring Green Campus

The first stage of the identification of the indicators was obtained from the study of selected assessment tools that were used to measure green HEIs globally. There are 9 chosen tools, and one of the tools is the UI Greenmetric (UIGM). The other tools education indicators were studied are 2) The Sustainability Assessment Questionnaire (SAQ), 3) Campus Sustainability Assessment Framework (CSAF), 4) Auditing Instrument for Sustainability in Higher Education (AISHE), 5) Sustainability Tracking, Assessment, and Rating System (STARS), 6) Adaptable Model for Assessing Sustainability (AMAS), 7) Sustainability Tool for Auditing for University Curricula in Higher Education (STAUNCH), 8) Graphical Assessment of Sustainability in Universities (GASU), and 9) Unit-based sustainability assessment tool (USAT). The reason for choosing these 9 SATs is that the tools were frequently mentioned in the study of (Dawodu et al., 2022; Du et al., 2020; Findler et al., 2018, 2019; Griebeler et al., 2021; Gutiérrez-Mijares et al., 2023; Husaini et al., 2018; Parvez & Agrawal, 2019; Singh et al., 2021; Zainordin & Ismail, 2018).

3.2 Stage 2 Data Collection for Questionnaire Survey

To validate the proposed indicators, a questionnaire survey was conducted at 9 of the most sustainable public HEIs ranked by UI Greenmetric in 2023, mainly targeted at the Faculty of Built Environment, Engineering or Architecture and any related faculty. This method was chosen based on a similar study of (Yaakub & Mohamed, 2020), which used a questionnaire survey. The faculty was selected due to the fact that a majority of the building assessors in Malaysia came from this background. However, academicians from other faculties were also accepted. A total population of 827 individuals was identified within the specific faculty, as detailed in **Table 2**. The sample size was determined based on a probability sampling technique, which involved random sampling. By using the Survey Monkey sample calculation method, with a standard confidence level of 95% with a 5% margin of error, a grand total of 262 academic samples is necessary. This approach is widely adopted in survey research to ensure the results are statistically reliable and generalizable to the target population.

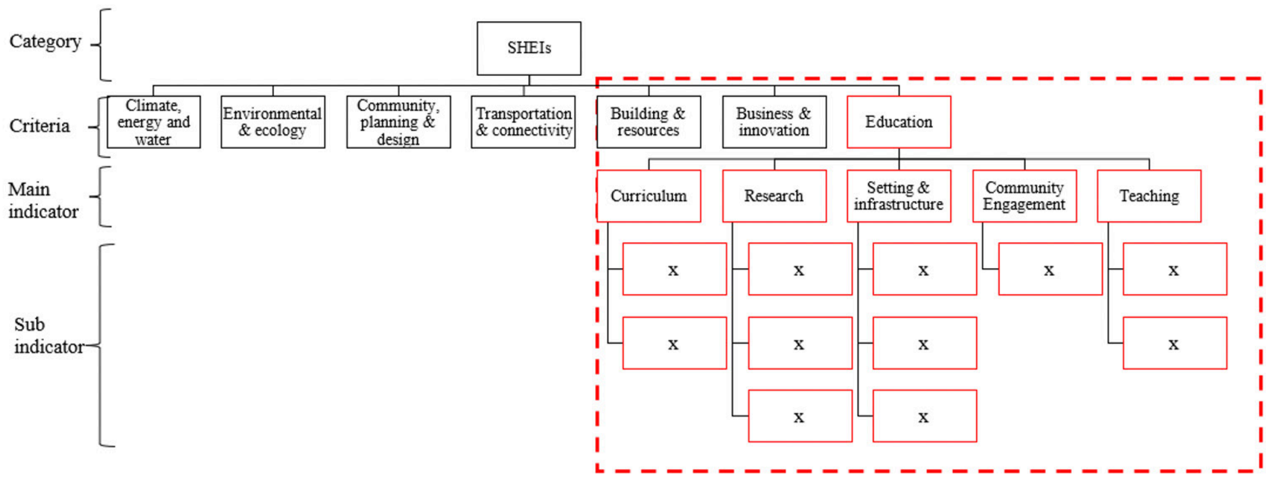

Figure 2: *Conceptual framework of the study*

Table 2: *Population size from selected faculty*

HEIs		Number of Academic Staff
UiTM	Faculty of Built Environment	275
UPM	Faculty of Forestry and Environment	76
UTM	Faculty of Built Environment & Surveying	167
UTHM	Faculty of Civil Engineering & Built Environment	139
UM	Faculty of Built Environment	78
UMP	Faculty of Civil Engineering Technology	75
UteM	Faculty Engineering	30
UNiSZA	Faculty of Innovative Design and Technology	32
UMS	Faculty Engineering	28
Grand Total		827

The questionnaire survey was divided into two sections: (1) Demographic background, and (2) Development of the education indicators. A five-point Likert scale was employed to assess the 30 indicators for measuring sustainable HEIs, with the scale values defined as follows: 1 for "strongly disagree," 2 for "disagree," 3 for "neither agree nor disagree," 4 for "agree," and 5 for "strongly agree." The Likert scale was chosen as one of the initiatives to minimize the potential bias in this study (Kusmaryono et al., 2022). The completion of the questionnaire will take around 15 to 20 minutes. Google Forms is chosen as the preferred way for delivering the questionnaire to respondents due to its user-friendly interface, simplicity, and comprehensive response data. The survey started on 12th August 2023 until 12th January 2024 and was distributed via emails, WhatsApp applications, and hard copy distribution. The first stage was conducted via email, beginning on 12th August 2023, as email is the most convenient way to distribute the survey (Stockemer & Stockemer, 2019). Some questionnaires were distributed via the WhatsApp application as an alternative but only for those who are reachable. Email reminders were sent after a few weeks to increase the percentage of response rate; however, after a few weeks of sending the questionnaire, the response rate was found to be very low (10% response rate). To boost the response rate, the hardcopy questionnaire was distributed to the HEIs after two (2) months (starting on 12th October 2024), and two (2) weeks were given for the respondents to complete and return their questionnaire. The face-to-face survey was supported by little incentive to increase the participant rates, which has been very effective in increasing participation, as suggested by Stockemer and Stockemer (2019). A total of five (5) months were taken as outlined in **Figure 3** to complete the questionnaire survey to collect 181 respondents due to the different locations of HEIs.

3.3 Data Analysis

In the initial stage of secondary data collection, a content analysis method was employed to systematically examine the data. All educational indicators from the selected assessment tools were extracted, categorized, and meticulously recorded in an Excel spreadsheet for further analysis and refinement. Meanwhile, the second stage of the data collection was recorded in two (2) formats: Google Form and hardcopy. Both types of data were compiled into an Excel Spreadsheet before the statistical analysis of the study was performed using the statistical software, SPSS. The data analysis was conducted using statistical methods based on a five-point Likert scale. The selection of statistical methods of data analysis was similar to the study of Leal et al. (2024).

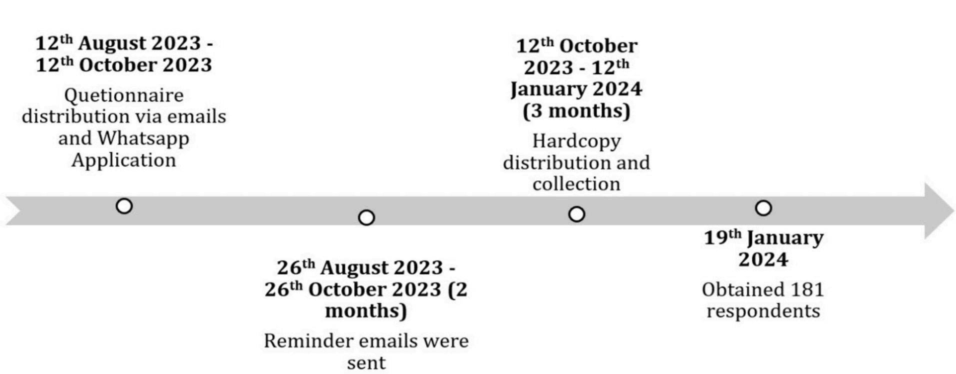

Figure 3: *Timeline for questionnaire survey*

3.4 Limitations of the Study

The first limitation of the study is its focus on the education indicators in developing sustainable HEIs, since the number of sustainable HEIs in Malaysia is lacking due to the absence of the education indicator in measuring sustainable HEIs (Parvez & Agrawal, 2019). Secondly, this study was conducted at nine (9) selected public HEIs in Malaysia. The chosen number is deemed appropriate, considering that the focus and limitation of the study are only towards public HEIs in Malaysia, where the total number of public HEIs was 20. The remaining 595 institutions, which include colleges, polytechnics, community colleges, and private institutions, fall outside the scope of the researcher's study. Nine (9) public HEIs were selected in this study, following a similar approach by Yaakub and Mohamed (2020) who selected 7 public HEIs and focused on either at the Faculty of Built Environment, Engineering or Architecture, which is another limitation of the study. The faculty was selected due to the fact that a majority of the building assessors in Malaysia came from this background. However, academicians from other faculties were also accepted.

4. RESULTS AND DISCUSSION

4.1 Stage 1 Results Regarding Identification of Education Indicators

The education indicators were studied based on the definition of sustainable HEIs that was pointed out by the previous experts (Aleixo et al., 2018; Atici et al., 2021; Biancardi et al., 2023; Dawodu et al., 2022; Du et al., 2020; Fissi et al., 2021; Freidenfelds et al., 2018; Gutiérrez-Mijares et al., 2023; Khovrak, 2020; Menon & Suresh, 2020; Rinawiyanti et al., 2023). Based on the definition of green or sustainable HEIs, the education aspect of green HEIs encompasses five key criteria: curriculum, research and innovation, campus setting and infrastructure, community engagement, and teaching. A total of 85 educational indicators were initially identified through the selection of SATs and subsequently cross-checked for relevance. After this rigorous evaluation, a refined set of 30 education indicators was finalized in **Table 3**, thereby fulfilling the first objective of the study.

The primary indicator identified was the curriculum, with its sub-indicators encompassing sustainability aspects related to courses, university programs, learning outcomes, educational initiatives, and funding for program development. The second key indicator was research, which integrates sustainability into various aspects such as research activities, publications, laboratory projects, collaborations, conferences, and research funding. These two indicators are consistently emphasized in the definition of green HEIs and are regarded as fundamental in the development of sustainable higher education institutions. The remaining criteria that meet the requirements of green HEIs include campus setting and infrastructure, which integrate sustainability into reports, websites, social media, guidelines, and assessments. The fourth indicator, community engagement, encompasses activities involving staff and students, cultural initiatives, event funding, corporate social responsibility efforts, student and staff organizations, and related start-ups. The final indicator, teaching, includes orientation programs, ongoing training, as well as funding for training and professional development.

Table 3: *Proposed education indicators*

Criteria	Education				
Indicators	Curriculum	Research	Setting & Infrastructure	Community Engagement	Teaching
Sub-indicators	1. Sustainability courses, 2. Sustainability University Program, 3. Sustainability Learning Outcome, 4. Sustainability focused-educational program, and 5. Funding for Program Development	6. Sustainable research, 7. Scholarly sustainable publication, 8. Laboratory projects, 9. Open access to research, 10. Collaboration, 11. Conferences, 12. Funding for research and innovation	13. Sustainable report, 14. Sustainable website 15. Sustainable media social, 16. Green Education Guideline, 17. Sustainable assessment	18. Student activities, 19. Staff activities, 20. Cultural activities, 21. Sustainable community services, 22. Funding for sustainable events, 23. Funding for corporate social responsibility, 24. Sustainability staff organization, 25. Sustainability student organization, 26. Sustainability related start-up	27. Orientation training, 28. On-going training, 29. Funding for training, 30. Funding for professional development

4.2 Stage 2 Result Addressing Demographics Profile of the Respondents

A total of 181 respondents obtained from the study represent 69% of the response rate. The analysis of the questionnaire survey is structured into two parts. The first part outlines the demographic background of the respondents. The second part is the required indicators to measure sustainable HEIs in Malaysia. To ensure the reliability of the education indicators, the questionnaire survey was distributed to the academicians at the public HEIs regarding their opinion on the education indicators to measure sustainable HEIs in Malaysia. This was achieved by determining the frequency of occurrences of education indicators from the respective respondents.

Section A asks general questions about the respondent's background. The purpose of this question was to help the researcher identify the academic background, teaching experience, the respondents' institutions, and involvement in campus sustainability of the respondents. The result is depicted in **Table 4**. The findings indicate that the lowest response rate, at 2.2%, was from a building surveying background, while academicians with a civil engineering background exhibited the highest response rate at 11.6% from a particular background. The highest percentage also comes from other backgrounds, which is 54.1% from real estate, park and amenity management, environmental management, forestry, social science, and many more. The other percentage of 3.9% of the respondents were Architecture background, 7.7% were mechanical and electrical engineering, 10.5% were quantity surveying, 6.1% were construction management, and 3.9% were estate management. This implies that there are two groups of academicians, almost half of the respondents have construction backgrounds, and the other half come from different areas of study. In terms of their academic background, the respondents, on average, have over five years of experience in the academic field. However, only 13.3% of the participants have less than five years of work experience. The data analysis showed that the respondents have substantial experience in the academic field and have shared their insights over the years, helping to shape the development of educational indicators. In summary, the relative experience and competence of the survey participants are sufficient and can be trusted to accurately represent the actual population, thus lending credibility to the collected survey data. Among the respondents, the highest percentage, 41.1%, was affiliated with UTHM, followed by the second highest number, 34.3% were from the UiTM. Meanwhile, the lowest responses, or none, were recorded from UMP and UniSZa. The respondents were inquired about their engagement in campus sustainability, including integrating sustainable learning outcomes into teaching and learning, conducting sustainable research, and developing sustainable programs. Findings revealed that 91.2% of respondents were actively involved in campus sustainability initiatives, while 8.8% were disengaged. This outcome indicates that the respondents were aware of and familiar with campus sustainability initiatives.

Table 4: *Demographic profile of respondents*

Area	Frequency (No)	Percentage (%)
Academic Background of Respondents		
Architecture	7	3.9
Civil Engineering	21	11.6
Mechanical and Electrical Engineering	14	7.7
Quantity Surveying	19	10.5
Building Surveying	4	2.2
Construction Management	11	6.1
Estate Management	7	3.9
Others	98	54.1
Academic experience		
More than ten years	128	70.7
Five years to 9 years	29	16.0
Less than five years	24	13.3
Higher Education Institutions		
UiTM	62	34.3
UPM	27	14.9
UTM	13	7.2
UTHM	75	41.4
UM	1	0.6
UMP	0	0
UteM	1	0.6
UNiSZA	0	0
UMS	2	1.1
Involvement in campus sustainability		
Yes	165	91.2
No	16	8.8

4.3 Stage 2 Result Focusing on Education Indicators in Measuring Green HEIs

The first objective of the study was achieved based on the review of nine selected SATs with a focus on education indicators. 30 key education indicators have been identified as essential for inclusion in sustainable assessment tools to evaluate green HEIs in Malaysia. All indicators were subsequently validated through a questionnaire survey. A five-point Likert scale was employed to assess the 30 indicators for measuring sustainable HEIs, with the scale values defined as follows: 1 for "strongly disagree," 2 for "disagree," 3 for "neither agree nor disagree," 4 for "agree," and 5 for "strongly agree." Indicators with mean scores ranging from 4.00 to 5.00 are considered to have a high frequency of occurrence. Those with mean scores between 3.00 and 3.99 are classified as having a moderate frequency of occurrence, while indicators with mean scores below 3.00 are considered to have a low frequency of occurrence.

As shown in **Table 5**, 30 indicators should be included in the education criteria for measuring green HEIs in Malaysia. The table presents the ranking of these education indicators, as rated by academicians from selected public HEIs in Malaysia. These developed education indicators contributed to achieving the second objective of this study. Following the interpretation of the five-point Likert scale, the analysis of the survey data revealed that the mean scores for the 30 education indicators ranged from 4.0497 to 4.3778, with standard deviations between 0.61822 and 0.79223. As a result, all 30 indicators were deemed appropriate for inclusion as education indicators in measuring sustainable HEIs, as factors with mean scores between 4.00 and 5.00 are considered to have a high frequency of occurrence, as outlined earlier. The analysis identified the top five key indicators for developing sustainable HEIs as: Funding for Program Development, Funding for Research Development, Funding for Training, Sustainable Courses, and Sustainable Research. These findings underscore the critical role of funding as a foundational criterion in advancing sustainability within HEIs alongside the integration of sustainability into academic curricula and research initiatives.

Based on the results, the adopted 30 indicators under the five main indicators shall be included in the education criteria to measure sustainable HEIs in Malaysia. The analysis identified the top five key indicators for developing sustainable Higher Education Institutions (HEIs) as: Funding for Program Development, Funding for Research Development, Funding for Training, Sustainable Courses, and Sustainable Research. These findings underscore the critical role of funding as a foundational criterion in advancing sustainability within HEIs alongside the integration of sustainability into academic curricula and research initiatives.

Table 5: *Proposed education indicators in measuring sustainability in higher education institutions*

Indicators	Mean	Standard deviation (SD)	Ranking
Funding for program development	4.3778	0.64432	1
Funding for research and development	4.3702	0.70000	2
Funding for training	4.3425	0.69427	3
Sustainable course	4.3425	0.64447	4
Sustainable research	4.3260	0.72176	5
Green education guideline	4.3260	0.69828	6
Funding for corporate social responsibility	4.3260	0.65730	7
Sustainable community service	4.3149	0.66270	8
Funding for professional development	4.3094	0.67772	9
On-going training	4.3039	0.66786	10
Scholarly sustainable publication	4.2762	0.72337	11
Sustainable assessment	4.2762	0.67572	12
Student Activities	4.2597	0.61822	13
Laboratory projects	4.2541	0.73903	14
Sustainable media social	4.2541	0.67622	15
Funding for sustainable events	4.2486	0.70637	16
Sustainable report	4.2431	0.73524	17
Sustainable website	4.2431	0.68841	18
Open access to research	4.2376	0.79856	19
Sustainable learning outcome	4.2155	0.73256	20
Sustainability focused - educational program	4.2155	0.72494	21
Sustainable university program	4.2155	0.71723	22
Staff activities	4.2099	0.67504	23
Conference	4.2044	0.65590	24
Sustainability student organizations	4.1823	0.69516	25
Orientation training	4.1713	0.68997	26
Sustainability related start-up	4.1547	0.68988	27
Collaborations	4.1492	0.79223	28
Sustainability staff organizations	4.1436	0.69228	29
Cultural activities	4.0497	0.70140	30

Beyond financial support, the presence of a sustainable curriculum and research indicators is essential for nurturing a new generation of students and academicians who are not only knowledgeable but also actively engaged in sustainability practices. This dual emphasis on teaching and research reinforces the institution's commitment to long-term sustainability goals.

These insights align with widely accepted definitions of sustainable HEIs, as proposed by numerous scholars, which emphasize that a truly sustainable institution must go beyond environmental protection to include curriculum transformation and research innovation as integral components of its sustainability framework (Fissi et al., 2021; Biancardi et al., 2023).

The developed education indicator facilitated the accomplishment of objective two in this study. Following the interpretation of the five-point Likert scale, the analysis of the survey data indicated that the mean score for the thirty (30) education indicators ranged from 4.0497 to 4.3778, and the standard deviation ranged from 0.61822 to 0.79223. This resulted in all thirty (30) indicators being agreed to be inserted as education indicators in measuring sustainable HEIs. This is because factors with mean scores between 4.00 and 5.00 are considered to have high-frequency occurrence indicators, as mentioned in previous paragraphs.

5. CONCLUSION

The need to establish a tool to measure sustainable HEIs is crucial, as the percentage of sustainable HEIs in the country is low, and collective efforts are needed to achieve the Sustainable Development Goals (SDGs). Among many types of HEIs, public and private HEIs are particularly important for a transition towards sustainable development. The research aims to propose the education indicators for measuring green HEIs in Malaysia. The sustainability knowledge needs to be embedded in HEIs either through curriculum, research, teaching and learning, etc. To achieve the study's aim, the first objective was accomplished through a questionnaire survey conducted at nine (9) selected public HEIs in Malaysia, with the data collection process taking five (5) months to complete. The second objective of the study was achieved through descriptive data analysis using the SPSS software and thirty (30) important indicators in measuring green HEIs. The top five indicators were identified as: (1) funding for program development, (2) funding for research and innovation, (3) funding for training, (4) sustainable courses, and (5) sustainable research. Funding was identified as the most important indicator to adopt sustainability in HEIs. It is agreed that a specific tool shall be developed to measure green HEIs in Malaysia, and all thirty (30) indicators for the education criteria shall be included in measuring green HEIs in Malaysia. The top three indicators identified were funding for program development, research and innovation, and training. This validates the viewpoint by Leal Filho et al. (2018), who argue that sustainable HEIs should start with creating sustainable programs and research and innovation activities, making financial support in these areas essential. Additionally, funding for training was necessary to support sustainable training initiatives. Other significant indicators emerged once funding for program development and sustainable research was initiated. The other required twenty-five (25) indicators can be acknowledged from the development of HEIs sustainable websites and sustainable social media like Facebook, Instagram, TikTok and many more, which are also part of the indicators in measuring sustainable HEIs. To support the future development of SATs, it is recommended that education indicators be simplified to encourage all HEIs to adopt sustainable practices. Additionally, a post-assessment is necessary for sustainable HEIs to ensure that sustainability efforts extend beyond the campus.

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